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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B.Tech (CBCS) Degree Examinations

May 2025

IS3301-1 – NoSQL DATABASE

(For IS)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **Why was NoSQL developed?**
A) To improve relational databases
B) To handle large-scale distributed data
C) To replace SQL completely
D) To store images only
2. **Document-oriented databases typically store data in:**
A) Rows and Columns
B) JSON or BSON format
C) Key-Value pairs
D) XML files
3. **Master-Slave replication ensures:**
A) Faster querying
B) High availability
C) Only one master node handles all writes
D) All nodes can handle writes
4. **Update consistency ensures:**
A) A single version of truth for writes
B) The database is always available
C) Replication delays are avoided
D) Replication delays are not avoided
5. **Which of the following NoSQL databases uses a graph structure?**
A) Cassandra
B) Neo4j
C) MongoDB
D) Redis
6. **In NoSQL databases, what does "schema-less" mean?**
A) No structure for data
B) Flexible structure for data
C) Data is always unstructured
D) Data has a predefined schema
7. **What is the role of a "combiner" in MapReduce?**
A) Merging intermediate outputs locally on the same node
B) Partitioning data across nodes
C) Aggregating results globally
D) Assigning tasks to nodes
8. **What type of database is MongoDB?**
A) Relational Database
B) NoSQL Database
C) Graph Database
D) Column-Oriented Database
9. **Which command is used to view all databases in MongoDB?**
A) show databases
B) list databases
C) db.show()
D) get databases
10. **In MongoDB, what is equivalent to a Row in RDBMS?**
A) Collection
B) Table
C) Document
D) Column
11. **How does MongoDB handle NULL values?**
A) By ignoring NULL fields
B) By replacing NULL with default values
C) By explicitly storing NULL values
D) By throwing an error

12. What does the findOne() method do?
 A) Returns all matching documents
 B) Returns only the first matching document
 C) Returns the last matching document
 D) Returns an array of matching documents
13. Which aggregation stage is used for grouping documents?
 A) \$match
 B) \$group
 C) \$sort
 D) \$filter
14. What is the purpose of an index in MongoDB?
 A) To store additional documents
 B) To improve query performance
 C) To compress the database
 D) To secure the database
15. What type of architecture does Cassandra use?
 A) Master-Slave
 B) Peer-to-Peer
 C) Single-Server
 D) Multi-Master
16. What is the primary query language used in Cassandra?
 A) SQL
 B) CQL
 C) NoSQL Query
 D) HiveQL
17. Which clause is used to define replication in a keyspace?
 A) WITH REPLICATION
 B) USING REPLICATION
 C) SET REPLICATION
 D) DEFINE REPLICATION
18. Which tool is used to import data into Cassandra?
 A) cqlsh
 B) nodetool
 C) cassandra-loader
 D) pgAdmin
19. What command is used to export data from Cassandra?
 A) COPY
 B) EXPORT
 C) SAVE
 D) SEND
20. How do you execute multiple CQL statements as a single transaction?
 A) BATCH
 B) TRANSACTION
 C) MULTIQUERY
 D) ATOMIC

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the four main types of NoSQL databases and give examples of each.	8	L*2	1	1
	b) Explain why NoSQL databases were introduced. What are the key advantages of NoSQL over traditional relational databases?	8	L2	1	1
2.	a) What is update consistency and read consistency in NoSQL databases? Provide examples of situations where each is important.	6	L1	1	1
	b) Explain why it is difficult to achieve all three properties of the CAP theorem simultaneously.	5	L2	2	1
	c) Explain sharding, and how does it work in NoSQL databases?	5	L2	2	1
3.	a) Explain the process of partitioning and combining in MapReduce. Why is it crucial for handling large datasets?	8	L2	2	1
	b) Explain the differences between Master-Slave Replication and Peer-to-Peer Replication.	8	L2	2	1
Unit – II					
4.	a) List and explain the different data types in MongoDB.	5	L1	3	1
	b) What is MongoDB? Explain its characteristics and why it is used?	5	L2	3	1

- c) Write MongoDB queries for the following:
- Insert a document into the students collection to store basic information about a student, including their roll number, name, grade, and hobbies.
 - Write a query to display the names and grades of all students who have a grade greater than 'B'.
 - Write a query to delete a student document from the students collection using the roll number "S123".
- 6 L3 4 2
5. a) Explain the save() and remove() methods in MongoDB. Include examples of their usage.
- 6 L2 4 1
- b) Write MongoDB queries to perform the following operations on the employees collection:
- Insert a new employee document with employee_id, name, department, and salary.
 - Use the save() method to insert or update an employee document. If the employee with employee_id: "E002" exists, update their details; otherwise, insert a new document.
 - Update the salary of the employee with employee_id: "E003" to 5500.
 - Remove the document of the employee with employee_id: "E004" from the employees collection.
 - Retrieve all employees from the "Sales" department and display only their name and salary.
- 10 L3 4 2
6. a) Explain how MongoDB handles arrays in documents and provide examples of array-specific queries to manipulate or retrieve data.
- 6 L2 4 1
- b) Write MongoDB queries for the following operations on the employees collection:
- Count the total number of employees who have a salary greater than 5000.
 - Retrieve the first 6 documents from the employees collection, but only include name and salary fields.
 - Sort the employees by hire_date in descending order.
 - Skip the first 3 employees and display the next 5 documents sorted by name.
 - Count how many employees work in the "Sales" department and have a salary greater than 3000.
- 10 L3 4 2

Unit – III

7. a) Explain how CRUD operations work in Cassandra with suitable CQL queries.
- 5 L2 5 2
- b) Explain different CQL data types with examples.
- 5 L2 3 1
- c) Explain the key features of Apache Cassandra that make it a preferred NoSQL database?
- 6 L2 5 1
8. a) What is collection? Explain set, List and Map collections.
- 5 L2 3 1
- b) How can we modify an existing table schema in Cassandra using ALTER commands? Provide an example.
- 5 L3 5 2
- c) Explain import and export operations in Cassandra with example.
- 6 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
